

LLM-Advisor: An LLM Benchmark for Cost-efficient Path Planning across Multiple Terrains

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Abstract—Multi-terrain cost-efficient path planning is a crucial task in robot navigation, requiring the identification of a path from the start to the goal that not only avoids obstacles but also minimizes travel costs. This is especially crucial for real-world applications where robots need to navigate diverse terrains in outdoor environments, where recharging or refueling is difficult. However, there is very limited research on this topic. In this paper, we develop a prompt-based approach, **LLM-Advisor**, which leverages large language models (LLMs) as **effective advisors for path planning**. The LLM-Advisor selectively provides suggestions, demonstrating its ability to recognize when no modifications are necessary. When suggestions are made, 70.59% of the paths suggested for the A* algorithm, 69.47% for the RRT* algorithm, and 78.70% for the LLM-A* algorithm achieve greater cost efficiency. Since LLM-Advisor may occasionally lack common sense in their suggestions, we propose two hallucination-mitigation strategies. Furthermore, we experimentally verified that GPT-4o performs poorly in zero-shot path planning, even when terrain descriptions are clearly provided, demonstrating its low spatial awareness. We also experimentally demonstrate that using an LLM as an advisor is more effective than directly integrating it into the path-planning loop. Since LLMs may generate hallucinations, using LLMs in the loop of a search-based method (such as A*) may lead to a higher number of failed paths, demonstrating that our proposed LLM-Advisor is a better choice.

I. INTRODUCTION

Path planning is a task of finding a path from a starting point to a goal while meeting specific conditions such as avoiding obstacles, minimizing travel distance or time, and adhering to other constraints [1]. This task is crucial in multiple fields such as robotics, autonomous navigation, industrial automation, and virtual environment navigation, as it directly impacts the efficiency, safety, and feasibility of operational systems [2]. Many path planning methods have been proposed in recent years, and they can effectively accomplish the planning task while ensuring the validity of the path in most cases [3]–[8].

Existing methods for path planning are mainly categorized into search-based and sampling-based approaches. Typical search-based methods include Breadth-First Search (BFS) [9], Depth-First Search (DFS) [10], Best-First Search [11], Dijkstra’s algorithm [12], A* [13], and various A* variants [14]–[20]. On the other hand, typical

sampling-based methods include Rapidly-exploring Random Tree (RRT) [21], RRT* [22], Informed RRT* [23], RRT-Connect [24], Dynamic RRT [25], Improved RRT [26], etc. However, these methods lack knowledge of the global environment, and they often generate paths with unnecessary detours, which may not be optimal.

Meanwhile, Large Language Models (LLMs) have a powerful capability for understanding global contexts [27]–[30], which is especially beneficial in complex problem-solving domains such as path planning [31]–[34]. These models have demonstrated the ability of LLMs to handle and reason over long-context inputs, providing global insights that reflect their understanding of the environment, such as identifying obstacles, agents, and the relative positions of goals. However, Ref. [35] found that LLMs perform poorly when dealing with complex, long-term planning and spatial reasoning tasks such as grid-based path planning. LLMs often generate invalid or impractical paths, leading to incomplete routes or collisions, indicating a capability gap when handling detailed spatial complexity. This paper aims to solve the above issues. We claim that by carefully designing the prompts, LLMs can serve as effective advisors for complex path-planning tasks.

In this paper, we first introduce a new task setting: cost-efficient path planning across multiple terrains, which is crucial for unmanned robots. While existing methods mainly consider the travel distance and time, the travel cost is also very important. Robots often operate in environments where recharging or refueling is challenging, such as remote areas, space, or disaster zones. Minimizing energy consumption allows robots to operate for longer periods without the need to recharge, thereby increasing their operational efficacy. Secondly, we propose an innovative path-planning method, LLM-Advisor. LLM-Advisor serves as a post-processing module for the path planning algorithms by leveraging global insights provided by large language models. It suggests improvements to the planned path when the optimal cost-efficient route has not been found. We experimentally found that: 1) By carefully designing the prompts, the proposed LLM-Advisor selectively provides suggestions only when necessary, demonstrating its ability to discern when adjustments are needed and when they are not, and 2) By integrating the proposed hallucination-mitigation strategies, the LLM-Advisor significantly improves its performance in suggesting paths. 3) When suggestions are made, 70.59% of the paths suggested for the A* algorithm, 69.47% for the RRT* algorithm, and 78.70% for the LLM-A* algorithm prove to be more cost-efficient. The contributions of this

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TABLE I
TERRAIN DIFFICULTY LEVELS, TRAVERSAL COSTS, AND
CORRESPONDING COLORS

Difficulty Level	0	1	2	3	4	5	6	7	8	9	10	11
Cost	1	3	3.5	4	4.5	5	5	6	6.5	7	8	∞
Color												

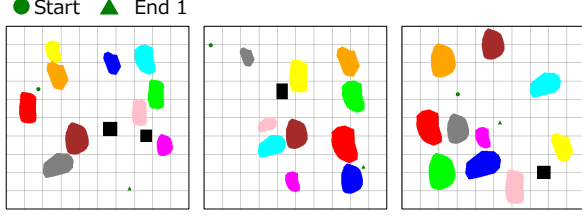


Fig. 1. A visualization of samples from the created MultiTerraPath dataset

paper are threefold:

- We propose a prompt-based method, LLM-Advisor, that enables LLMs to act as effective advisors after path planning is completed by other methods. Experimental results demonstrate the effectiveness of the proposed prompt and the validity of using LLMs as advisors for path planning.
- We propose two hallucination mitigation strategies to further enhance the performance of LLM-Advisor.
- Experimental results demonstrate that LLM-Advisor is capable of selectively making suggestions. Specifically, 70.59% of the paths suggested for the A* algorithm, 69.47% for the RRT* algorithm, and 78.70% for the LLM-A* algorithm are more cost-efficient.

II. METHOD

In this section, we first introduce the dataset used in this paper. Then, we describe the details of the proposed LLM-Advisor.

A. Datasets

MultiTerraPath Dataset. This paper introduces a synthetic dataset for evaluating model performance, MultiTerraPath. Our MultiTerraPath dataset contains 1,000 maps, each sized 500×500 . Each map includes several high-cost terrains and one or two impassable obstacles, with one start-to-end pair defined per image. The dataset can be used for tasks such as cost-efficient path planning and obstacle-avoidance path planning. The defined terrains, traversal costs, and their corresponding colors in the images are summarized in Table I. Fig. 1 presents some sample maps from the dataset. The coordinates of the start and end points of each pair are saved in the corresponding JSON file.

RUGD.v2 Dataset. The RUGD dataset [36] supports semantic understanding of unstructured outdoor scenes, including rough and challenging terrains. There are four scenes in the RUGD dataset: creek, park, trail, and village. The size of each image is 688×555 . Based on the RUGD dataset, we create one start-to-end pair per image, forming our

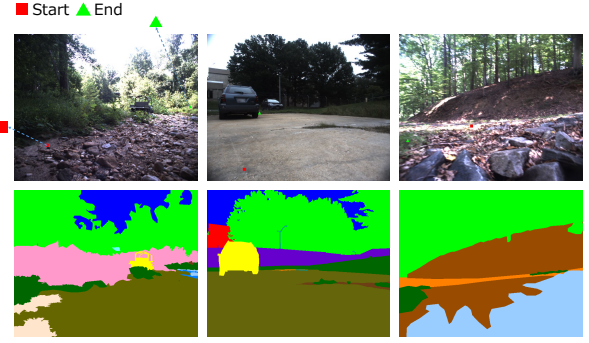


Fig. 2. A visualization of samples from the created RUGD.v2 dataset

RUGD.v2 dataset. The traversal costs for different terrains are synthetically defined by GPT-4o (Table III). Fig. 2 shows some examples from the dataset. Note that in real-world applications, it is crucial to conduct thorough experiments to accurately determine the traversal costs for each terrain type. These experiments help to fine-tune the cost values by considering factors such as robot types and the physical properties of the terrain.

B. LLM-Advisor

The motivations for designing LLM-Advisor are twofold: 1) Existing path planning methods, whether search-based or sampling-based, focus on local regions when conducting path planning. This ensures a reasonable path but may result in unnecessary detours, which significantly affect the total traversal cost. 2) Although some LLM-powered path planning methods have already been developed [34], [37], Ref. [37] leveraged an LLM in an interactive manner, which requires several rounds conversations. Therefore, it is not suitable for real-time operations. Ref. [34] leveraged an LLM to enhance the efficiency of the A* algorithm by generating a candidate list during the search process. However, the LLM-A* approach [34] led to an increased rate of failures in finding valid paths. Considering the above two issues, we propose using LLMs as advisors as a viable solution. However, due to the LLM's low spatial awareness and tendency for hallucinations, careful prompt design and the implementation of mitigation strategies are essential.

Main Prompt. Prompt design is a key component of the proposed LLM-Advisor. Specifically, when designing the main prompt, the terrain description is included as knowledge, as it provides critical information about the path planning environment. To address the issue of low spatial awareness, we also incorporate a detailed description of the planned path. By asking the LLMs to evaluate both the terrain and the path, we aim to enhance their spatial awareness and improve its performance. The code in Fig. 3 shows the designed prompt, where the GPT-4o is utilized. Examples of the terrain descriptions and detailed path descriptions used are presented in Fig. 4.

Hallucination-mitigation Strategies. Furthermore, we propose two mitigation strategies to alleviate the hallucinations of LLM-Advisor. Details are given in Fig. 5.

TABLE II
THE DEFINITIONS OF TRAVERSAL COSTS FOR THE RUGD_v2 DATASET

Color												
Label	Rock	Grass	Sign	Water	Pole	Tree	Bicycle	Gravel	Concrete	Log	Fence	Rockbed
Traversal Cost	∞	1.8	∞	∞	∞	∞	∞	1.5	1	∞	∞	3.0

Color												
Label	Bridge	Dirt	Table	Mulch	Sky	Person	Building	Container (generic object)	Alphalt	Bush	Sand	Vehicle
Traversal Cost	∞	2.2	∞	2.5	∞	∞	∞	∞	1.2	∞	3.5	∞

You are provided with a terrain cost grid description: {terrain_description} You are also provided with the real coordinates of start and goal points, and a description of path list computed by A*. Below is the data for each pair:
****Start-End Pair ****: - Start: {start} - End: {end} - A* path: {path_description}
Analyzing the A* paths, evaluate whether the most cost-efficient path has already been found for each pair.
IMPORTANT: **If not, you must suggest a path list, ensure the first point is the exact start coordinates and the last point is the exact end coordinates as provided, display every single coordinate pair.**
IMPORTANT: **Make your reply strictly in the following format:**
Requirements: If a path is optimal, the answer should be Yes. If a path is not optimal, the answer should be No, the suggested path coordinates (P). The list of coordinates (P) must start at the corresponding start coordinate and end at the corresponding end coordinate. The output format must strictly follow the format below, without any additional information. Output Format:
Yes or No, No or [P]
IMPORTANT: Only output the real list of coordinates for P. Do not include any extra explanations or text.

Fig. 3. The designed prompt for LLM-Advisor

Terrain description: High-cost area with cost 5 is approximately located between grid coordinates (42, 110) and (354, 454). High-cost area with cost 4 is approximately located between grid coordinates (53, 268) and (131, 407). High-cost area with cost 6 is approximately located between grid coordinates (98, 113) and (350, 224). High-cost area with cost 3 is approximately located between grid coordinates (148, 38) and (264, 430). High-cost area with cost 8 is approximately located between grid coordinates (200, 346) and (271, 383). High-cost area with cost 7 is approximately located between grid coordinates (349, 102) and (413, 183). Obstacles are approximately located between grid coordinates (268, 270) and (312, 395).

Path description (Detailed): Point 1 at (172, 88) has a terrain cost of 1. Point 2 at (173, 89) has a terrain cost of 1. Point 3 at (174, 90) has a terrain cost of 1. Point 4 at (175, 91) has a terrain cost of 1. Point 5 at (176, 92) has a terrain cost of 1. Point 6 at (177, 93) has a terrain cost of 1. Point 7 at (178, 94) has a terrain cost of 1. Point 8 at (179, 95) has a terrain cost of 1. ... Point 285 at (437, 337) has a terrain cost of 5. Point 286 at (438, 337) has a terrain cost of 5. Point 287 at (439, 337) has a terrain cost of 5. Point 288 at (440, 337) has a terrain cost of 5. Point 289 at (441, 337) has a terrain cost of 5. Point 290 at (442, 337) has a terrain cost of 5. Point 291 at (443, 337) has a terrain cost of 5. The total cost of the path is 410.

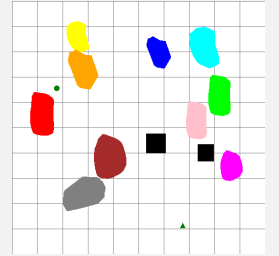


Fig. 4. Examples of terrain descriptions and detailed path descriptions for the A*-planned path in map 1 of the MultiTerraPath dataset

- **DescPath.** Since LLM-Advisor relies heavily on the contextual understanding capabilities of LLMs, directly generating coordinate points of the path may introduce hallucinations. Therefore, we propose generating a descriptive path (DescPath) instead of just listing individual path coordinates.
- **RAG.** We also propose a retrieval-augmented strategy (RAG) to mitigate these hallucinations. Specifically, for the MultiTerraPath dataset, we randomly retrieve one path planning example from the entire dataset. For the RUGD_v2 dataset, we randomly select a path-planning example from the same scene and integrate the retrieved

Retrieval-augmented strategy: You are provided with a terrain cost grid description based on historical tasks: {retrieved_terrain_description}. Additionally, you have the start and goal coordinates along with computed A-star paths retrieved from similar tasks: {retrieved_path_description}

Descriptive output: Requirements: If the path is optimal, respond with “Yes.” If the path is not optimal, respond with “No,” followed by a descriptive sentence indicating that the suggested path is not feasible, and then provide an alternative path. The description should clearly outline the movement from the start coordinate to the end coordinate, step by step. The output must strictly follow the format below, without any additional information. **Output Format:** Yes or No, No or from $(start_x, start_y)$ it goes to (x_1, y_1) , then (x_2, y_2) , ..., finally, it arrives at (end_x, end_y) .

Fig. 5. Hallucination-mitigation strategies

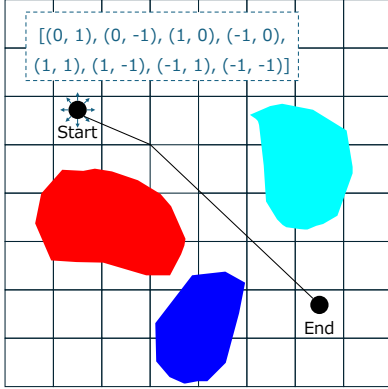


Fig. 6. The A* algorithm search eight neighbors

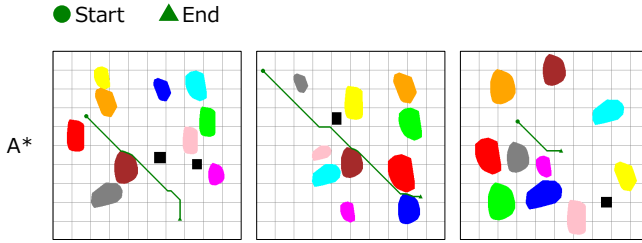


Fig. 7. Visualization of the A* searched paths

information into the original prompt.

III. EXPERIMENTAL RESULTS

A. Experimental Setup and Evaluation Metrics

We utilize a GPT-4o to implement our proposed prompt method. We applied the proposed LLM-Advisor to check the planned path generated by A* [13], RRT* [22], and LLM-A* [34]. A* is a search-based method, RRT* is a sampling-based method, and LLM-A* is an LLM-based method. Note that the experiments with LLM-A* [34] also use a GPT-4o. To further evaluate the effectiveness of the LLM-Advisor, we also compare it with LLM-A* [34] and GPT-4o.

Overview of A*. A* is a search-based path planning algorithm. It evaluates the eight neighboring points around the current position and determines the most cost-efficient path by selecting the next step with the lowest combined cost of traversal and estimated remaining distance(Fig. 6).

Therefore, A* generates reasonable paths, as illustrated in Fig. 7. However, the generated path often includes unnecessary detours. Although A* avoids obstacles, the overall path may not be the most cost-efficient.

Overview of RRT*. RRT* is a sampling-based path planning algorithm that extends RRT [21]. It randomly samples points and connects each new point to the nearest neighbor in the tree. Unlike RRT, RRT* evaluates the cost from all nearby nodes and selects the lowest-cost connection. After connecting, it rewires the tree whenever a more efficient path is found. Over time, this iterative rewiring process improves the overall path quality. Although time-consuming, RRT* generates reasonable paths, which is why we use our LLM-Advisor to evaluate its results, aiming to assess its effectiveness as a validation tool for precomputed reasonable paths.

Overview of LLM-A*. LLM-A* leverages LLMs to provide a candidate list for the A* algorithm. The main idea of LLM-A* is to improve search efficiency in obstacle-avoidance path planning. By restricting the search to the candidate list provided by LLM, LLM-A* can significantly decrease the search time. The prompt used for experiments using LLM-A* [34] is given in Fig. 8.

Since LLM-Advisor functions as an advisor, we assess its performance using two metrics: Relative Precision (RP) and Improvement Ratio (IR). The definition of RP is provided in Eq. 1, and the definition of IR is given in Eq. 2. RP represents the proportion of improved paths within the suggestions made by LLM-Advisor, while IR indicates the percentage of paths that are equal to or better than the original ones after incorporating LLM-Advisor. IR can reveal how many paths are valid and effective.

$$RP = \frac{N_{\text{Improved}}}{N_{\text{Suggested}}}, \quad (1)$$

where N_{Improved} denotes the number of improved paths and $N_{\text{Suggested}}$ denotes the number of suggested paths.

$$IR = \frac{N_{\text{Improved}} + N_{\text{Equal}}}{N_{\text{Path}}}, \quad (2)$$

where N_{Equal} denotes the number of paths that are identical to the A*-planned path, and N_{Path} represents the total number of planned paths.

You are provided with a terrain cost grid description: {terrain.description}
 You are also provided with the real coordinates of start and goal points,
 Start-End Pair: - Start: {start} - End: {end}
 Suggest a list of intermediate target states, P, to help an A* algorithm plan the most cost-efficient path. The output format must strictly follow the format below, without any additional information. Output Format: [P]
 IMPORTANT: Only output the real list of coordinates for P. Do not include any extra explanations or text.

Fig. 8. LLM-A* Prompt

TABLE III

EVALUATION OF PLANNED PATHS ACROSS DIFFERENT METHODS ON THE MULTITERRAPATH DATASET USING THE PROPOSED LLM-ADVISOR

Method	Terrain description	Path description Detailed	Brief	N_{Images}	N_{Path}	$N_{\text{Suggested}}$	N_{Improved}	RP (%) $\uparrow$
LLM-Advisor + A* [13]	✓	✓	-	1,000	1,000	612	432	70.59
LLM-Advisor + RRT* [22]	✓	✓	-	1,000	1,000	737	512	69.47
LLM-Advisor + LLM-A* [34]	✓	✓	-	1,000	1,000	770	606	78.70

TABLE IV

EVALUATION OF PLANNED PATHS ACROSS DIFFERENT METHODS ON THE RUGD_v2 DATASET USING THE PROPOSED LLM-ADVISOR

Method	Scenes	N_{Images}	N_{Path}	$N_{\text{Suggested}}$	N_{Improved}	RP (%) $\uparrow$
LLM-Advisor + A*	Creek	1000	1000	876	580	66.21
	Park	1000	1000	860	720	83.72
	Trail	1000	1000	620	540	87.10
	Village	1000	1000	780	560	71.79

TABLE V

COMPARISON OF OUR LLM-ADVISOR WITH OTHER LLM-BASED METHODS

Method	Terrain description	N_{Images}	N_{Path}	$N_{\text{Successful}}$	N_{Improved}	N_{Equal}	$N_{\text{Deteriorated}}$	IR (%) $\uparrow$	FPS $\uparrow$
A* [13]	-	1,000	1,000	1,000	-	-	-	-	9.39
GPT-4o	✓	1,000	1,000	1,000	208	0	792	20.80	1.21
LLM-A* [34]	✓	1,000	1,000	938	152	538	248	<u>69.00</u>	3.10
LLM-Advisor + A*	✓	1,000	1,000	1,000	432	393	175	82.50	<u>5.86</u>

B. Main Results

Main Comparisons. Table III presents the main comparison results for the MultiTerraPath dataset. Most suggestions made by the proposed LLM-Advisor are more cost-efficient. It achieves an RP of 70.59% when making suggestions for the A*-planned path, 69.47% for the RRT*-planned path, and 78.70% for the LLM-A*-planned path. This demonstrates that, with detailed path descriptions, LLM-Advisor can effectively serve as a reliable validation tool for planned paths generated by other methods. Moreover, the proposed LLM-Advisor is versatile. It can improve the previous approaches regardless of whether they are search-based, sampling-based, or LLM-based. Table IV shows experimental results for the the RUGD_v2 dataset.

Table V presents a comparison of LLM-Advisor+A* with A*, LLM-A* [34] and GPT-4o. The best performance for each evaluation metric is bolded, while the second-best is underlined. A* successfully generated valid paths for all cases. Since the proposed LLM-Advisor functions

as an advisor, evaluating the paths generated by A*, it also successfully validated all paths. GPT-4o successfully planned all paths, although most were suboptimal or invalid. However, it was still able to generate a path. In LLM-A [34], GPT-4o suggests a target list, providing a sequence of intermediate target states for A* to consider during path planning. If the generated target list contains points within the obstacle region, the neighbor search process cannot proceed. As shown in Table VI, there were 62 paths that were not successfully planned using LLM-A*. When considering the Improvement Ratio (IR), LLM-Advisor+A* achieved a higher IR, indicating better performance. The GPT-4o has the lowest IR, indicating that GPT-4o does not have ability to do zero-shot path planning. Moreover, since our LLM-Advisor acts as a postprocessor, it is possible to involve humans in the verification process, allowing us to decide whether to accept the paths suggested by the LLM-Advisor. Incorporating human judgment in this process can further improve the quality of the planned paths.

To evaluate the computational efficiency of different path

TABLE VI
ABLATION STUDIES

Method	Terrain description	Path description Detailed	Brief	RAG	Descriptive output	N_{Images}	N_{Path}	$N_{Suggested}$	$N_{Improved}$	RP (%) $\uparrow$
LLM-Advisor + A*	✓	-	✓	-	-	1,000	1,000	602	130	21.59
	✓	✓	-	-	-	1,000	1,000	607	352	57.99
	✓	✓	-	✓	-	1,000	1,000	610	413	67.70
	✓	✓	-	-	✓	1,000	1,000	609	399	65.52
	✓	✓	-	✓	✓	1,000	1,000	612	432	70.59

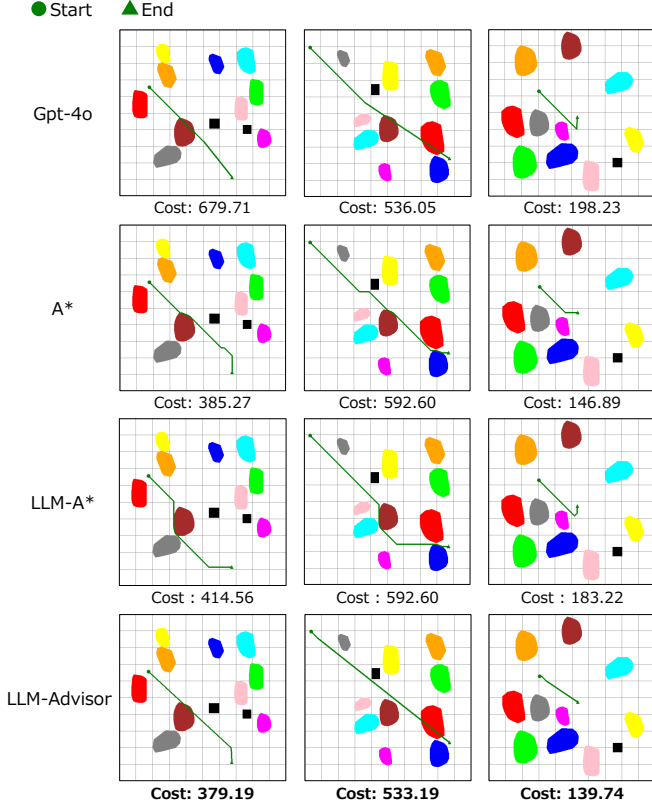


Fig. 9. Visualization of the planned paths by GPT-4o, A*, LLM-A*, and LLM-Advisor + A*. Black bold numbers indicate the favorable suggestions, while red bold numbers highlight less favorable suggestions. All other numbers represent cases where no suggestions were made. Lower costs indicate better performance

planning methods, we report frames per second (FPS) for each approach in Table V. FPS quantifies the number of paths processed per second, where a higher value indicates greater efficiency. All methods are executed entirely on the CPU. Among them, LLM-Advisor + A* selectively makes suggestions only when an improvement is needed, minimizing unnecessary LLM computations and achieving a higher FPS compared to other LLM-based methods.

Visualizations. Fig. 9 presents some path planning results. As shown in the figure, our LLM-Advisor successfully suggests more cost-efficient paths. Fig. 10 provides visualization results for the RUGD_v2 dataset.

Ablation Studies. Table VI shows some ablation study results. We evaluate LLM-Advisor with both brief and detailed

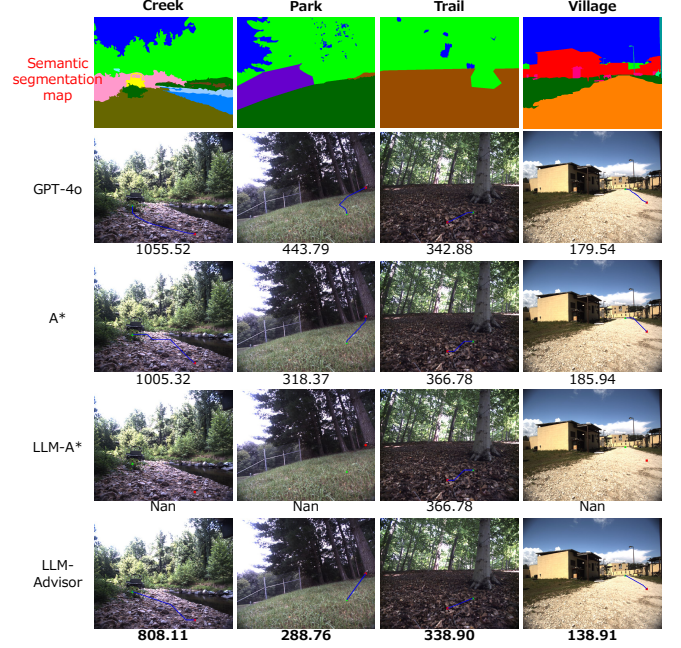


Fig. 10. Visualization of experimental results on RUGD_v2 dataset

path descriptions to demonstrate the effectiveness of the proposed prompt. One example of brief path description is given in Fig. 11. As shown, the level of detail in path descriptions significantly impacts final performance. By using detailed path descriptions, the LLM-Advisor can more accurately evaluate the A*-planned path and make a higher-quality suggestion. We also found that it has only a limited influence on the number of suggestions. This indicates that the ability of LLM-Advisor to suggest improvements to planned paths is not affected much by the path descriptions. Moreover, using the proposed hallucination-mitigation strategies, the LLM-Advisor achieves higher RP. Fig. 12 shows the visualization results.

IV. FUTURE WORK

The proposed LLM-Advisor operates as a zero-shot model, relying heavily on the performance of general-purpose pretrained LLMs. Without task-specific training, it leverages the inherent knowledge and reasoning capabilities of these models to provide guidance and insights. However, its performance may be limited when dealing with domain-specific challenges. As a next step, we plan to fine-tune

Path description (Brief): From (88, 172) to (305, 370) the terrain cost is 1. From (306, 371) to (306, 371) the terrain cost is 9223372036854775807. From (307, 372) to (337, 413) the terrain cost is 1. From (337, 414) to (337, 443) the terrain cost is 5. The total cost of the path is -9223372036854775399.

Fig. 11. Path description (Brief)

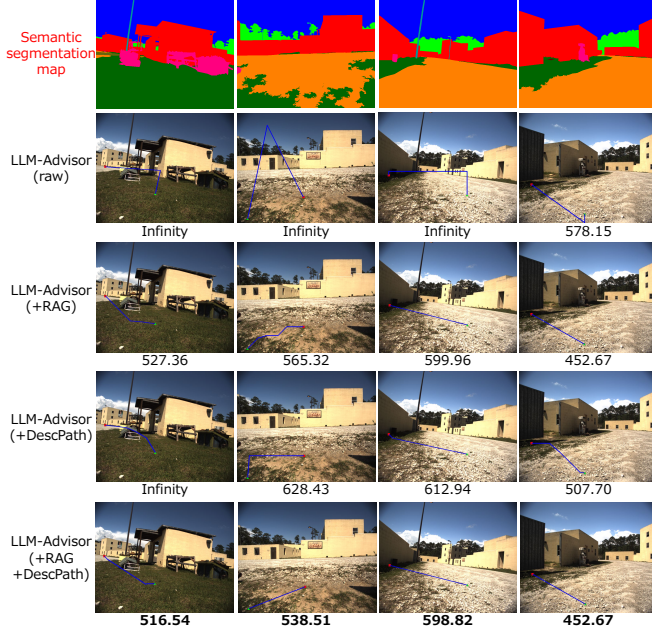


Fig. 12. Visualization of experimental results after integrating the proposed hallucination-mitigation strategies

the model using task-relevant data to enhance its accuracy, adaptability, and effectiveness in path planning and related tasks.

V. CONCLUSIONS

This paper introduced a novel prompt-based approach, LLM-Advisor, which achieves high relative precision and improvement ratio in evaluating paths planned by existing methods. The LLM-Advisor is versatile and can be used to verify paths generated by any planing method. It provides suggestions when existing methods fail to identify the most cost-efficient path. Furthermore, we propose two strategies to mitigate hallucinations. Our experiments on created MultiTerraPath and RUGD_v2 datasets demonstrate the effectiveness of LLM-Advisor. We also experimentally found that GPT-4o struggles with effective path planning based solely on terrain descriptions. We believe that our work offers valuable insights for the path planning community, particularly for researchers investigating the integration of LLMs into path planning.

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